

# ESGData.xlsx

## Practical Data Dictionary & Scoring Guide

S&P 500 sustainability index - updated dataset review

|                 |                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------|
| <b>Workbook</b> | 3 sheets: General_Features, Scope_Time_Series, Financial_Time_Series                              |
| <b>Universe</b> | 503 listed securities representing 500 unique issuer names                                        |
| <b>Purpose</b>  | Define each parameter, assess data quality, and state how it should/should not enter an ESG score |
| <b>Version</b>  | 13 September 2026                                                                                 |

Primary source: uploaded ESGData.xlsx. Bloomberg field mnemonics are preserved exactly. Where the workbook does not store unit metadata, the document flags the unit for terminal FLDS verification rather than guessing.

# 1. Executive assessment

**Verdict:** the updated workbook is materially better and is now strong enough to support a credible hackathon ESG score. The new social-policy and governance variables substantially improve balance across E/S/G. However, the score should remain selective: many variables are sparse, redundant, non-monotonic, or nearly constant.

| Area                   | What is strong                                                  | Main limitation                                                                | Recommended treatment                                                                          |
|------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Environmental          | Excellent 2020-2024 carbon coverage; energy is broad            | Water/waste/event fields are sparse; current snapshot has mixed/latest vintage | Use annual FY2024 carbon + sector-adjusted resource intensities; sparse events as penalties    |
| Social                 | Women workforce 90.9%, management 65.2%; useful safety outcomes | Safety coverage 30-41%; many policy fields are disclosure/process signals      | Outcome metrics first; combine policy Y/N fields into one small policy block                   |
| Governance             | Most fields >90% coverage                                       | Several fields have almost zero variance or no clear monotonic direction       | Use independence, CEO duality, attendance, executive diversity; drop constant/redundant fields |
| Financial denominators | Revenue/assets/headcount coverage is very high                  | Ordinary financial performance is not sustainability                           | Use only for normalization, exposure and controls                                              |

## Critical data-quality findings

- 503 securities but 500 issuer names. Alphabet, Fox and News Corp each appear twice via share classes. Collapse to issuer level before ranking.
- The General\_Features GHG snapshot matches FY2025 Scope 1/2/3 in roughly 94.2% / 94.6% / 93.9% of comparable rows. It is therefore mostly 2025/latest, but not guaranteed to be a uniform fiscal-year snapshot.
- FY2024 annual GHG coverage is strong: Scope 1 92.8%, Scope 2 90.9%, Scope 3 84.1%. FY2025 is much less complete: 67.2%, 63.8%, 57.5%. Use FY2024 for the comparable cross-sectional carbon score.
- Scope\_Time\_Series contains a duplicated GHG\_SCOPE\_1\_2019 column where Scope 2 2019 should be. The values exactly repeat Scope 1 2019. Do not use 2019 Scope 2 until re-extracted.
- TOTAL\_GHG\_EMISSIONS is redundant with Scope 1 + Scope 2 in 274/277 complete rows. Never count all three at full weight.
- AUDIT\_CMTE\_INDEPENDENCE\_FLD\_SCR is 10 for every observed company (498 rows), and ESG\_LINKED\_COMPENSATION\_FOR\_BRD is N for every observed company (419 rows). Both currently provide no ranking information.

## Recommended missing-data rule

**Never replace Bloomberg N/A with zero.** Distinguish observed zero from missing. Compute each sub-score over applicable observed features, calculate coverage/confidence separately, and shrink low-coverage scores toward 50 if you need a single robust ranking:

$$\text{Adjusted score} = 50 + \text{Coverage} \times (\text{Raw score} - 50)$$

#N/A N/A = unavailable/not reported in the pull. #N/A Field Not Applicable = Bloomberg flags the field as not applicable; for SBTi this should be cross-checked with the official target dashboard before interpreting absence.

## Scoring-role legend

|      |           |                 |              |              |         |
|------|-----------|-----------------|--------------|--------------|---------|
| CORE | SECONDARY | ADVERSE PENALTY | CLIMATE ONLY | CONTEXT / QC | EXCLUDE |
|------|-----------|-----------------|--------------|--------------|---------|

## 2. Identifiers and peer-group variables

| Field                    | Definition                                                     | Unit / encoding | Coverage | Role       | Direction | Practical preprocessing / caveat                                                                               |
|--------------------------|----------------------------------------------------------------|-----------------|----------|------------|-----------|----------------------------------------------------------------------------------------------------------------|
| ID                       | Bloomberg security identifier / ticker used in the workbook.   | Text            | 100.0%   | JOIN ONLY  | -         | Use for joins inside the workbook; do not score. Share classes create multiple securities for the same issuer. |
| name()                   | Company/issuer name returned by Bloomberg.                     | Text            | 100.0%   | JOIN ONLY  | -         | Use with ID/ISIN for entity resolution. 500 unique issuer names across 503 securities.                         |
| id_isin()                | ISIN of the listed security.                                   | Text            | 100.0%   | JOIN ONLY  | -         | Use for security-level joins; not issuer aggregation. Different share classes have different ISINs.            |
| gics_sector_name()       | GICS sector classification.                                    | Category        | 99.8%    | PEER GROUP | -         | Use for sector neutralization and materiality rules. One row is missing a sector in this pull.                 |
| GICS_INDUSTRY_GROUP_NAME | GICS industry-group classification; more granular than sector. | Category        | 100.0%   | PEER GROUP | -         | Preferred peer group when enough firms are available; fall back to sector. 25 industry groups are present.     |

Coverage is measured on 503 security rows in General\_Features.

**Issuer aggregation rule:** score the company once. For the three duplicate issuers, ESG values are identical across share classes; retain one issuer record and map both securities back to the same company score.

### 3. Environmental and climate variables

| Field               | Definition                                                                                                       | Unit / encoding                               | Coverage | Role                    | Direction | Practical preprocessing / caveat                                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------|-------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GHG_SCOPE_1         | Direct GHG emissions from sources owned or controlled by the company.                                            | 000 metric tonnes CO2e                        | 58.4%    | CORE                    | Lower     | Use FY2024 from Scope_Time_Series for cross-sectional scoring; divide by revenue; log1p before peer ranking. Current snapshot is mostly FY2025/latest and has lower coverage. |
| GHG_SCOPE_2         | Indirect GHG emissions associated with purchased energy.                                                         | 000 metric tonnes CO2e                        | 55.1%    | CORE                    | Lower     | Combine with Scope 1 for operational carbon intensity; use consistent year. Accounting method (market/location based) is not encoded in this workbook.                        |
| GHG_SCOPE_3         | Other indirect value-chain GHG emissions upstream and downstream.                                                | 000 metric tonnes CO2e                        | 49.3%    | CORE - LOWER WEIGHT     | Lower     | Score separately from Scope 1+2; intensity to revenue; sector-adjust; winsorize/log. Coverage and methodology are less comparable across companies.                           |
| TOTAL_GHG_EMISSIONS | Reported total operational GHG emissions. In this workbook it is effectively Scope 1 + Scope 2, not Scope 1+2+3. | 000 metric tonnes CO2e                        | 55.3%    | FALLBACK ONLY           | Lower     | Use only if Scope 1/2 components are unavailable; never double count with them. Matches Scope1+Scope2 in 274 of 277 complete rows.                                            |
| ENERGY_CONSUMPTION  | Total energy consumed during the reporting period.                                                               | Numeric; unit metadata not stored in workbook | 87.7%    | CORE / SECONDARY        | Lower     | Normalize by revenue or activity; log; sector peer ranking. Confirm Bloomberg FLDS unit before publication.                                                                   |
| RENEW_ENERGY_USE    | Amount of energy consumed that comes from renewable sources.                                                     | Typically 000 MWh; confirm FLDS               | 35.6%    | CORE AS RATIO           | Higher    | Derive renewable share = RENEW_ENERGY_USE / ENERGY_CONSUMPTION; clip to [0,1] after unit validation. Do not score raw renewable use; large firms mechanically use more.       |
| WATER_CONSUMPTION   | Total water consumed by the company.                                                                             | Numeric; unit metadata not stored in workbook | 37.6%    | SECONDARY / MATERIALITY | Lower     | Use water intensity (water/revenue) and only in water-material sectors or sector-relative scoring. Low-to-medium coverage; mixed reporting boundaries.                        |
| PCT_WATER_RECYCLED  | Percentage of water use supplied from recycled/reused water sources.                                             | %                                             | 5.8%     | IF TIME                 | Higher    | Use directly after validation; peer rank. Coverage is only about 6%, so exclude from main score.                                                                              |
| DISCHARGE_TO_WATER  | Reported discharge to water / water-pollution quantity.                                                          | Numeric; verify FLDS                          | 1.0%     | REJECT MAIN SCORE       | Lower     | Only use as an event/controversy flag if available. ~1% coverage makes it unusable for a broad ranking.                                                                       |
| TOTAL_WASTE         | Total waste generated/disposed during the reporting period.                                                      | Typically 000 metric tonnes; confirm FLDS     | 36.8%    | SECONDARY               | Lower     | Use waste/revenue; log; sector rank. Absolute quantities are size-biased.                                                                                                     |
| HAZARDOUS_WASTE     | Amount of hazardous waste discarded/generated.                                                                   | Typically 000 metric tonnes                   | 26.0%    | SECONDARY               | Lower     | Prefer hazardous-waste intensity or hazardous share of total waste. Do not double count heavily with total waste.                                                             |
| WASTE_RECYCLED      | Amount of waste recycled.                                                                                        | Typically 000 metric tonnes                   | 28.6%    | SECONDARY AS RATIO      | Higher    | Derive recycled share = WASTE_RECYCLED / TOTAL_WASTE when units/boundaries align. Raw recycled tonnage is size-biased.                                                        |
| NUM_ENVIRON_FINES   | Number of environmental fines disclosed for the reporting period.                                                | Count                                         | 7.4%     | ADVERSE PENALTY         | Lower     | Use as a one-sided controversy penalty; log1p count. Missing is not zero.                                                                                                     |
| ENVIRON_FINES_AMT   | Total amount of environmental fines paid/disclosed during the period.                                            | Millions of currency units; verify currency   | 8.3%     | ADVERSE PENALTY         | Lower     | Normalize by revenue and apply one-sided penalty. Very sparse; missing is not evidence of no fines.                                                                           |
| NUMBER_SPILLS       | Number of accidental spills disclosed in the reporting period.                                                   | Count                                         | 5.4%     | ADVERSE PENALTY         | Lower     | Use as one-sided event penalty; sector-relative. Very sparse and materiality differs strongly by industry.                                                                    |
| AMOUNT_OF_SPILLS    | Total reported quantity of spills.                                                                               | Typically 000 tonnes; verify FLDS             | 4.2%     | ADVERSE PENALTY         | Lower     | Log/normalize and use only when disclosed. Very sparse.                                                                                                                       |

| Field                                | Definition                                                                                 | Unit / encoding | Coverage | Role              | Direction       | Practical preprocessing / caveat                                                                                                                                                                |
|--------------------------------------|--------------------------------------------------------------------------------------------|-----------------|----------|-------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PCT_OF_GREEN_SUSTAIN_REVENUE</b>  | Percentage of company revenue/turnover classified by the company as green or sustainable.  | %               | 2.0%     | REJECT MAIN SCORE | Higher          | Could be a side statistic only. Only ~2% coverage and company-defined taxonomy create comparability risk.                                                                                       |
| <b>SBTI_NEAR_TERM_TARGET_STATUS</b>  | Status of the company's near-term target in the Science Based Targets initiative universe. | Category        | 51.3%    | CLIMATE ONLY      | Ordered         | Suggested order for transition ambition: Targets Set > Committed > Removed; verify N/A against SBTi directly. Do not treat absence as bad performance until applicability/matching is verified. |
| <b>SBTI_NEAR_TERM_TARGET_CLASSES</b> | Temperature ambition/classification of the validated near-term SBTi target.                | Category        | 43.7%    | CLIMATE ONLY      | Higher ambition | Example order: 1.5C > Well-below 2C > 2C; keep a small weight relative to realized emissions. Use official SBTi data if time permits; classification conventions have evolved.                  |

Coverage is measured on 503 security rows in General\_Features.

## 4. Social variables

| Field                       | Definition                                                                       | Unit / encoding                     | Coverage | Role                  | Direction              | Practical preprocessing / caveat                                                                                                                                              |
|-----------------------------|----------------------------------------------------------------------------------|-------------------------------------|----------|-----------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UN_GLOBAL_COMPACT_SIGNATORY | Whether the company is a signatory to the UN Global Compact.                     | Y/N                                 | 57.9%    | LOW-WEIGHT POLICY     | Y positive             | Encode Y=1, N=0; combine with other policy indicators in one block. Process/commitment signal, not an outcome.                                                                |
| HUMAN_RIGHTS_POLICY         | Whether the company explicitly discloses a human-rights policy.                  | Y/N                                 | 77.3%    | SECONDARY POLICY      | Y positive             | Binary; combine with policy block rather than score independently at large weight. High prevalence; susceptible to boilerplate disclosure.                                    |
| EQUAL_OPPORTUNITY_POLICY    | Whether an equal-opportunity/non-discrimination policy is disclosed.             | Y/N                                 | 79.9%    | SECONDARY POLICY      | Y positive             | Binary; policy block. Very high prevalence, so little cross-sectional discrimination.                                                                                         |
| FAIR_REMUNERATION_POLICY    | Whether the company discloses a policy addressing fair remuneration.             | Y/N                                 | 57.1%    | SECONDARY POLICY      | Y positive             | Binary; policy block. Presence is not proof of wage outcomes.                                                                                                                 |
| POLICY_AGAINST_CHILD_LABOR  | Whether the company discloses a policy against child labor.                      | Y/N                                 | 70.4%    | SECONDARY POLICY      | Y positive             | Binary; policy block. More material in exposed supply chains; sector materiality matters.                                                                                     |
| WORK_ACCIDENTS_EMPLOYEES    | Number of employee work accidents reported during the period.                    | Count                               | 30.4%    | CORE SAFETY           | Lower                  | Normalize per 1,000 employees; log/peer-rank. Raw count is dominated by workforce size.                                                                                       |
| FATALITIES_CONTRACTORS      | Number of contractor fatalities related to company sites/facilities/operations.  | Count                               | 18.3%    | SEVERE PENALTY        | Lower                  | Use one-sided severe penalty; do not reward zero excessively when reporting is uncertain. Avoid double counting with FATALITIES_TOTAL.                                        |
| FATALITIES_TOTAL            | Total reported fatalities across employees/contractors where reported.           | Count                               | 18.3%    | CORE / SEVERE PENALTY | Lower                  | Primary fatality variable if reliably defined; normalize by workforce/exposure where possible. Do not score alongside both employee and contractor fatalities at full weight. |
| EMPLOYEE_TRAINING_COST      | Amount spent on employee training during the reporting period.                   | Currency amount; typically millions | 8.9%     | IF TIME               | Contextually higher    | Normalize per employee; log; use small weight. Spend is an input, not proof of training quality; currency must be consistent.                                                 |
| COMMUNITY_SPENDING          | Company spending/contributions directed to communities/philanthropic programs.   | Currency amount; typically millions | 39.0%    | IF TIME               | Contextually higher    | Normalize by revenue; keep very low weight or report separately. Can reflect philanthropy rather than core business sustainability.                                           |
| EMPLOYEE_TURNOVER_PCT       | Employee turnover as a percentage of the workforce.                              | %                                   | 28.2%    | CORE / ASYMMETRIC     | Lower when excessive   | Score high turnover as adverse relative to sector; avoid assuming zero turnover is optimal. Strong sector/labor-model effects.                                                |
| PCT_WOMEN_EMPLOYEES         | Women as a percentage of the total workforce.                                    | %                                   | 90.9%    | CONTEXT / DIVERSITY   | Balance                | Use mainly to contextualize management representation; do not simply reward the maximum percentage. A balanced workforce is not a monotonic 'more is always better' metric.   |
| PCT_WOMEN_MGT               | Women as a percentage of management.                                             | %                                   | 65.2%    | CORE DIVERSITY        | Higher / parity        | Use together with workforce share; derive management representation gap = women_mgt - women_employees. More decision-relevant than workforce share alone.                     |
| FATALITIES_EMPLOYEES        | Number of employee fatalities reported in the period.                            | Count                               | 34.6%    | SEVERE PENALTY        | Lower                  | Normalize per 100k employees if used; one-sided penalty. Prefer FATALITIES_TOTAL if both sources/boundaries align.                                                            |
| LOST_TIME_INCIDENT_RATE     | Rate of incidents resulting in lost work time, typically per 200,000 work hours. | Rate                                | 40.6%    | CORE SAFETY           | Lower                  | Use directly after winsorization and sector peer ranking. One of the best comparable safety outcomes in the dataset.                                                          |
| PCT_EMPLOYEES_UNIONIZED     | Percentage of employees represented by/affiliated with unions.                   | %                                   | 47.3%    | CONTEXT ONLY          | No universal direction | Report or use as contextual feature, not a positive/negative score. Unionization depends heavily on sector, country and labor model.                                          |

Coverage is measured on 503 security rows in General\_Features.

## 5. Governance variables

| Field                                     | Definition                                                                                                              | Unit / encoding | Coverage | Role                  | Direction                   | Practical preprocessing / caveat                                                                                                                     |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------|----------|-----------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| BOARD_SIZE                                | Number of directors serving on the board.                                                                               | Count           | 92.8%    | CONTEXT ONLY          | Non-monotonic               | Use as control or plausibility check, not a direct ESG score. Neither larger nor smaller is universally better.                                      |
| PCT_INDEPENDENT_DIRECTORS                 | Independent directors as a percentage of total board membership.                                                        | %               | 99.4%    | CORE GOVERNANCE       | Higher                      | Use directly/peer rank; winsorization usually unnecessary. High coverage and interpretable.                                                          |
| CEO_DUALITY                               | Whether the CEO also serves as board chair.                                                                             | Y/N             | 93.0%    | CORE GOVERNANCE       | N generally preferred       | Encode duality Y=1 as weaker separation of oversight, with a modest penalty. Governance norms differ by jurisdiction; keep weight moderate.          |
| AUDIT_CMTE_INDEPENDENCE_FLD_SCR           | Bloomberg audit-committee independence score, typically on a 0-10 scale.                                                | Score 0-10      | 99.0%    | EXCLUDE CURRENTLY     | Higher                      | Do not include in current ranking. All 498 observed values are 10, so it has zero cross-sectional information.                                       |
| SIZE_OF_AUDIT_COMMITTEE                   | Number of members on the audit committee.                                                                               | Count           | 92.8%    | CONTEXT ONLY          | Non-monotonic               | Use as diagnostic/control only. Size does not map cleanly to governance quality.                                                                     |
| BOARD_AVERAGE_TENURE                      | Average tenure of current board directors.                                                                              | Years           | 99.2%    | CONTEXT / OPTIONAL    | Non-monotonic               | If used, prefer a mid-range/sector-relative treatment, not 'higher is better'. Long tenure adds experience but may reduce refreshment/independence.  |
| BOARD_AVERAGE_AGE                         | Average age of board members.                                                                                           | Years           | 92.4%    | CONTEXT ONLY          | No universal direction      | Do not score monotonically. Age is not governance quality.                                                                                           |
| PCT_OF_NON_EXEC_DIR_ON_BOARD              | Non-executive directors as a percentage of total board membership.                                                      | %               | 92.8%    | REDUNDANT / SECONDARY | Higher                      | Use only if independent-director data are missing. Highly overlaps conceptually with board independence.                                             |
| CSR_SUSTAINABILITY_COMMITTEE              | Whether a board/management committee specifically addresses CSR/sustainability matters.                                 | Y/N             | 93.4%    | SECONDARY GOVERNANCE  | Y positive                  | Binary, modest weight. High prevalence; presence does not guarantee effectiveness.                                                                   |
| ESG_LINKED_COMPENSATION_FOR_BOARD         | Whether board compensation is explicitly linked to ESG criteria.                                                        | Y/N             | 83.3%    | EXCLUDE CURRENTLY     | Y positive                  | Do not use in current ranking; consider extracting executive ESG-linked bonus instead. All 419 observed values are N, so there is no discrimination. |
| PCT_NON_EXEC_DIR_ON_AUDIT_COMMITTEE       | Percentage of audit committee members who are non-executive directors.                                                  | %               | 99.2%    | EXCLUDE / QC          | Higher                      | Use as quality-control flag only. 497 of 499 observed values are 100; almost no variance.                                                            |
| PCT_NON_EXEC_DIR_ON_COMPNSATION_COMMITTEE | Percentage of compensation committee members who are non-executive directors.                                           | %               | 97.2%    | EXCLUDE / QC          | Higher                      | Use as quality-control flag only. 484 of 489 observed values are 100; almost no variance.                                                            |
| CHIEF_EXECUTIVE_OFFICER_TENURE            | Length of the current CEO's tenure.                                                                                     | Years           | 99.0%    | CONTEXT ONLY          | Non-monotonic               | Do not score monotonically; optional control. Long tenure can indicate experience or entrenchment.                                                   |
| PCT_OF_EXECUTIVES_THAT_ARE_WOMEN          | Women as a percentage of company executives / senior executive group.                                                   | %               | 87.3%    | CORE DIVERSITY        | Higher / parity             | Use as leadership-diversity outcome; consider parity-oriented transformation. Good coverage and more senior than workforce/management measures.      |
| BOARD_DURATION                            | Length of a single director board term before renewal; commonly 1 year for annual elections or 3 for classified boards. | Years           | 93.2%    | SECONDARY GOVERNANCE  | Shorter generally preferred | Treat 1-year annual election as more shareholder-accountable than 3-year classified term; modest weight. Distinct from BOARD_AVERAGE_TENURE.         |
| BOARD_MEETINGS_PER_YEAR                   | Number of board meetings held during the year.                                                                          | Count           | 92.4%    | CONTEXT ONLY          | Non-monotonic               | Do not reward more meetings mechanically. High meeting frequency can reflect active oversight or crisis.                                             |
| BOARD_MEETING_ATTENDANCE_PCT              | Percentage attendance at board meetings.                                                                                | %               | 89.5%    | CORE GOVERNANCE       | Higher                      | Use directly with a ceiling effect; penalize low attendance more than reward 100 vs 98. Interpretable governance participation metric.               |

Coverage is measured on 503 security rows in General\_Features.

## 6. Scope\_Time\_Series: annual emissions panel

**Purpose.** This sheet is the preferred source for the carbon-performance component because it provides explicit fiscal-year history. For the hackathon ranking, use FY2024 as the common current year and 2020-2024 for trend estimation. Treat FY2025 as an optional latest-update layer, not the common ranking base.

| Parameter family | Definition                              | Years      | Unit      | Recommended derived feature                                          | Use                      |
|------------------|-----------------------------------------|------------|-----------|----------------------------------------------------------------------|--------------------------|
| GHG_SCOPE_1_YYYY | Direct operational GHG emissions        | 2019-2025  | 000 tCO2e | Scope1/revenue; log-linear trend in absolute emissions and intensity | Core climate performance |
| GHG_SCOPE_2_YYYY | Purchased-energy indirect GHG emissions | 2020-2025* | 000 tCO2e | (Scope1+Scope2)/revenue; trend                                       | Core climate performance |
| GHG_SCOPE_3_YYYY | Value-chain emissions                   | 2019-2025  | 000 tCO2e | Scope3/revenue; separate trend with lower weight                     | Core, lower weight       |

\*The workbook currently has no valid Scope 2 2019 column: the would-be column is labeled GHG\_SCOPE\_1\_2019 and duplicates Scope 1 values.

### Exact annual fields and coverage

| Field                               | Coverage | Recommended status | Note                                                               |
|-------------------------------------|----------|--------------------|--------------------------------------------------------------------|
| GHG_SCOPE_1_2019                    | 80.1%    | OPTIONAL           | Can extend history; Scope 2 2019 is currently missing.             |
| GHG_SCOPE_1_2020                    | 86.7%    | HISTORY            | Use in trend estimation.                                           |
| GHG_SCOPE_1_2021                    | 91.5%    | HISTORY            | Use in trend estimation.                                           |
| GHG_SCOPE_1_2022                    | 92.4%    | HISTORY            | Use in trend estimation.                                           |
| GHG_SCOPE_1_2023                    | 92.6%    | HISTORY            | Use in trend estimation.                                           |
| GHG_SCOPE_1_2024                    | 92.8%    | PRIMARY            | Use as common snapshot.                                            |
| GHG_SCOPE_1_2025                    | 67.2%    | LATEST ONLY        | Incomplete reporting; do not rank all firms on this year.          |
| GHG_SCOPE_1_2019 [duplicate column] | 80.1%    | FIX                | Duplicate of Scope 1 2019; should be re-extracted as Scope 2 2019. |
| GHG_SCOPE_2_2020                    | 84.1%    | HISTORY            | Use in trend estimation.                                           |
| GHG_SCOPE_2_2021                    | 89.1%    | HISTORY            | Use in trend estimation.                                           |
| GHG_SCOPE_2_2022                    | 89.9%    | HISTORY            | Use in trend estimation.                                           |
| GHG_SCOPE_2_2023                    | 90.5%    | HISTORY            | Use in trend estimation.                                           |
| GHG_SCOPE_2_2024                    | 90.9%    | PRIMARY            | Use as common snapshot.                                            |
| GHG_SCOPE_2_2025                    | 63.8%    | LATEST ONLY        | Incomplete reporting; do not rank all firms on this year.          |
| GHG_SCOPE_3_2019                    | 69.0%    | OPTIONAL           | Can extend history; Scope 2 2019 is currently missing.             |
| GHG_SCOPE_3_2020                    | 73.4%    | HISTORY            | Use in trend estimation.                                           |
| GHG_SCOPE_3_2021                    | 78.9%    | HISTORY            | Use in trend estimation.                                           |
| GHG_SCOPE_3_2022                    | 81.7%    | HISTORY            | Use in trend estimation.                                           |
| GHG_SCOPE_3_2023                    | 84.3%    | HISTORY            | Use in trend estimation.                                           |
| GHG_SCOPE_3_2024                    | 84.1%    | PRIMARY            | Use as common snapshot.                                            |
| GHG_SCOPE_3_2025                    | 57.5%    | LATEST ONLY        | Incomplete reporting; do not rank all firms on this year.          |

### Recommended emissions model

- Current footprint:  $\text{FY2024 Scope 1+2 intensity} = (\text{Scope1} + \text{Scope2}) / \text{SALES\_REV\_TURN\_2024}$ .
- Value-chain footprint: FY2024 Scope 3 / revenue, kept separate and lower-weight because Scope 3 boundaries vary.
- Historical transition:  $\text{fit log(emissions intensity)} = \alpha + \beta \times \text{year over 2020-2024}$ ; more negative beta is better. Require at least 3 valid annual observations.
- Also keep an absolute-emissions trend: revenue growth can otherwise make intensity improve while total emissions still rise.
- SBTi should measure target credibility/ambition, not substitute for realized decarbonization.

## 7. Financial\_Time\_Series: denominators and controls

Do not turn these fields into an ESG pillar. They exist to normalize physical impacts, construct exposure-adjusted social metrics, and optionally control for company size/profitability.

| Parameter family             | Definition                                                       | Years     | Unit/encoding                                                         | Why it exists                                                                         | Score treatment               | Key caveat                                                                         |
|------------------------------|------------------------------------------------------------------|-----------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------|
| SALES_REV_TURN_YYYY          | Sales / revenue / turnover                                       | 2019-2025 | Scaled financial currency; workbook does not store currency metadata  | Primary denominator for emissions, energy, water, waste, fines and community spending | Use                           | Confirm currency override before comparing non-USD reporters.                      |
| BS_TOT_ASSET_YYYY            | Total assets                                                     | 2019-2025 | Scaled financial currency                                             | Alternative scale denominator for financials/real estate; size control                | Use selectively               | For banks/insurers, assets are more meaningful scale than revenue for some ratios. |
| EBITDA_YYYY                  | Earnings before interest, tax, depreciation and amortization     | 2019-2025 | Scaled financial currency                                             | Profitability control; possible denominator for spending                              | Control only                  | Missing for some financial companies; do not impute for ESG scoring.               |
| CF_CAP_EXPEND_PRPTY_ADD_YYYY | Capital expenditure / property additions in cash-flow convention | 2019-2025 | Scaled financial currency; mostly negative cash outflow in this sheet | Ordinary CAPEX control; use abs(value) if intensity is needed                         | Do not score as 'green capex' | Negative values are a sign convention, not negative investment.                    |
| NUM_OF_EMPLOYEES_YYYY        | Employee headcount                                               | 2019-2025 | Employees                                                             | Normalize accidents/fatalities/training; workforce scale control                      | Use                           | Headcount definitions can differ (average vs year-end; full-time vs total).        |

### Exact annual financial fields and coverage

| Field                        | Coverage | Primary use                              |
|------------------------------|----------|------------------------------------------|
| SALES_REV_TURN_2019          | 98.0%    | Emissions/resource intensity denominator |
| SALES_REV_TURN_2020          | 98.4%    | Emissions/resource intensity denominator |
| SALES_REV_TURN_2021          | 99.2%    | Emissions/resource intensity denominator |
| SALES_REV_TURN_2022          | 99.4%    | Emissions/resource intensity denominator |
| SALES_REV_TURN_2023          | 99.6%    | Emissions/resource intensity denominator |
| SALES_REV_TURN_2024          | 99.8%    | Emissions/resource intensity denominator |
| SALES_REV_TURN_2025          | 100.0%   | Emissions/resource intensity denominator |
| BS_TOT_ASSET_2019            | 97.6%    | Scale/control                            |
| BS_TOT_ASSET_2020            | 98.2%    | Scale/control                            |
| BS_TOT_ASSET_2021            | 98.4%    | Scale/control                            |
| BS_TOT_ASSET_2022            | 99.2%    | Scale/control                            |
| BS_TOT_ASSET_2023            | 99.4%    | Scale/control                            |
| BS_TOT_ASSET_2024            | 99.8%    | Scale/control                            |
| BS_TOT_ASSET_2025            | 100.0%   | Scale/control                            |
| EBITDA_2019                  | 91.7%    | Control / spending normalization         |
| EBITDA_2020                  | 91.8%    | Control / spending normalization         |
| EBITDA_2021                  | 92.8%    | Control / spending normalization         |
| EBITDA_2022                  | 92.8%    | Control / spending normalization         |
| EBITDA_2023                  | 92.6%    | Control / spending normalization         |
| EBITDA_2024                  | 93.0%    | Control / spending normalization         |
| EBITDA_2025                  | 93.4%    | Control / spending normalization         |
| CF_CAP_EXPEND_PRPTY_ADD_2019 | 98.0%    | Control only; abs() if used              |
| CF_CAP_EXPEND_PRPTY_ADD_2020 | 98.0%    | Control only; abs() if used              |
| CF_CAP_EXPEND_PRPTY_ADD_2021 | 99.0%    | Control only; abs() if used              |
| CF_CAP_EXPEND_PRPTY_ADD_2022 | 99.2%    | Control only; abs() if used              |
| CF_CAP_EXPEND_PRPTY_ADD_2023 | 99.2%    | Control only; abs() if used              |
| CF_CAP_EXPEND_PRPTY_ADD_2024 | 99.4%    | Control only; abs() if used              |
| CF_CAP_EXPEND_PRPTY_ADD_2025 | 99.8%    | Control only; abs() if used              |
| NUM_OF_EMPLOYEES_2019        | 91.7%    | Social safety/training denominator       |
| NUM_OF_EMPLOYEES_2020        | 93.6%    | Social safety/training denominator       |
| NUM_OF_EMPLOYEES_2021        | 94.2%    | Social safety/training denominator       |
| NUM_OF_EMPLOYEES_2022        | 95.6%    | Social safety/training denominator       |
| NUM_OF_EMPLOYEES_2023        | 97.2%    | Social safety/training denominator       |
| NUM_OF_EMPLOYEES_2024        | 97.0%    | Social safety/training denominator       |
| NUM_OF_EMPLOYEES_2025        | 96.4%    | Social safety/training denominator       |

## 8. Practical scoring blueprint using this workbook

**Recommended architecture:** transparent hierarchical peer percentiles, not PCA/K-means/neural aggregation.

| Output                   | Core ingredients                                                                                                                           | Treatment                                                                                 | What not to do                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Environmental score      | FY2024 Scope1+2 intensity; Scope3 intensity; energy intensity; renewable share; material water/waste features                              | Sector/industry percentiles; latest common year; adverse incidents as one-sided penalties | Do not add raw Scope1, Scope2 and TOTAL_GHG as separate equal features      |
| Social score             | LTIR; accident/fatality rates; women in management/executives; management representation gap; small policy block                           | Normalize counts by employees; sector adjust turnover/safety; cap policy contribution     | Do not reward unionization, community spending or policy count mechanically |
| Governance score         | Independent directors; CEO/chair separation; board attendance; executive gender representation; modest board-duration/CSR committee signal | Simple interpretable transforms; avoid redundant near-constant committee variables        | Do not score board size, age, meeting count or CEO tenure monotonically     |
| Climate Transition Score | 2020-2024 realized emissions trend + FY2024 footprint + SBTi status/ambition                                                               | Keep separate from ESG; realized performance should dominate target ambition              | Do not let a target compensate for rising emissions                         |

### Suggested derived variables

| Derived feature                | Formula                                                                                     | Use              |
|--------------------------------|---------------------------------------------------------------------------------------------|------------------|
| <b>S12_INTENSITY_2024</b>      | $(\text{GHG\_SCOPE\_1\_2024} + \text{GHG\_SCOPE\_2\_2024}) / \text{SALES\_REV\_TURN\_2024}$ | Core E / Climate |
| <b>S3_INTENSITY_2024</b>       | $\text{GHG\_SCOPE\_3\_2024} / \text{SALES\_REV\_TURN\_2024}$                                | Core E / Climate |
| <b>RENEWABLE_SHARE</b>         | $\text{RENEW\_ENERGY\_USE} / \text{ENERGY\_CONSUMPTION}$                                    | E                |
| <b>WASTE_INTENSITY</b>         | $\text{TOTAL\_WASTE} / \text{SALES\_REV\_TURN\_latest\_aligned}$                            | E                |
| <b>WASTE_RECYCLE_SHARE</b>     | $\text{WASTE\_RECYCLED} / \text{TOTAL\_WASTE}$                                              | E                |
| <b>ACCIDENTS_PER_1000_EMP</b>  | $1000 \times \text{WORK\_ACCIDENTS\_EMPLOYEES} / \text{NUM\_OF\_EMPLOYEES}$                 | S                |
| <b>FATALITIES_PER_100K_EMP</b> | $100000 \times \text{FATALITIES\_EMPLOYEES} / \text{NUM\_OF\_EMPLOYEES}$                    | S                |
| <b>MGMT_REPRESENTATION_GAP</b> | $\text{PCT\_WOMEN\_MGT} - \text{PCT\_WOMEN\_EMPLOYEES}$                                     | S                |
| <b>CLIMATE_TREND_S12</b>       | Slope of $\log((S1+S2)/\text{revenue})$ over 2020-2024                                      | Climate          |
| <b>DATA_COVERAGE</b>           | Weighted share of applicable score features observed                                        | Confidence       |

### Implementation order for a 24-hour hackathon

- BUILD NOW: issuer deduplication; missing-value parser; FY2024 carbon intensities; 2020-2024 emissions trends; core S/G outcomes; peer percentiles; coverage/confidence.
- IF TIME REMAINS: water/waste materiality rules; sparse fines/spills penalties; official SBTi refresh; robustness checks across alternative weights.
- REJECT FOR NOW: imputation models, PCA/autoencoders, green-revenue field, discharge-to-water, raw policy-count score, ordinary CAPEX as a sustainability signal.

## 9. Missing values, units and source notes

### Missing-value semantics

| Workbook value            | Interpretation                            | Scoring action                                                   |
|---------------------------|-------------------------------------------|------------------------------------------------------------------|
| Numeric zero              | Observed zero / reported zero             | Keep as a real observation.                                      |
| #N/A N/A                  | Value not available in the Bloomberg pull | Keep missing; do not convert to zero.                            |
| #N/A Field Not Applicable | Bloomberg marks field as not applicable   | Exclude from applicable denominator; verify SBTi cases directly. |
| Blank/None                | No value in workbook                      | Treat as missing unless extraction logic says otherwise.         |

### Unit caution

**The workbook stores values but not Bloomberg field-unit metadata.** GHG units are well established as thousands of metric tonnes CO2e. Several waste/spill/energy fields are commonly distributed with scaled units, but before the final presentation you should confirm each raw unit in Bloomberg FLDS/DAPL metadata. Ratios are only valid when numerator and denominator scaling is understood.

### Reference sources used to validate field interpretation

- Bloomberg Professional Services, Greenhouse Gas Emissions methodology: [https://assets.bbhub.io/professional/sites/10/1025089\\_CDS\\_REF\\_Greenhouse\\_Emissions\\_WP.pdf](https://assets.bbhub.io/professional/sites/10/1025089_CDS_REF_Greenhouse_Emissions_WP.pdf)
- Science Based Targets initiative - Target Dashboard: <https://sciencebasedtargets.org/target-dashboard>
- Bloomberg field names and definitions cross-checked against published academic data dictionaries using Bloomberg ESG mnemonics; Bloomberg FLDS remains authoritative for units and current definitions.

### Final recommendation

**Use this workbook as the core ESG source, but do not treat every column as a score input.** The strongest design is selective: outcome variables drive E/S/G, process/policy variables receive small weights, non-monotonic governance fields stay contextual, sparse adverse events act as penalties, and emissions history + SBTi form a separate Climate Transition Score.